

AI 2002: Artificial Intelligence

Introduction

Spring 2026

Textbook

- Stuart Russell and Peter Norvig, Artificial Intelligence. A Modern Approach, 4th Global Edition, Prentice Hall, Inc.
- Examination papers will be based solely on the textbook (not slides or other sources).

MARKS DISTRIBUTION (Tentative)

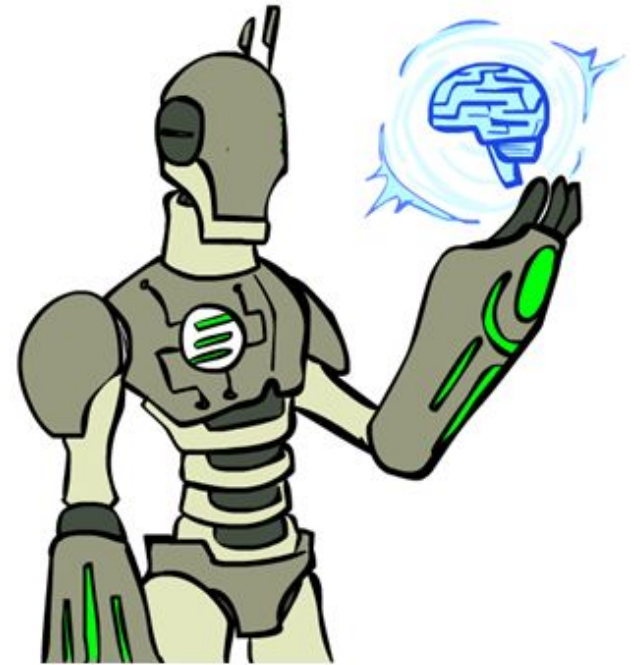
Final		50
Mid-Terms	2	30
Assignments	3 (weightage 3, 3, and 4)	10
Quiz	3 (weightage 3, 3, and 4)	10

Before: What is AI?

- Before: What is AI? What can it do? What is this course?

First Look at:

- What is the real-world impact of AI
- or
- 1.5 Risks and Benefits of AI



AI is having real-world impact

- Public imagination
 - Text assistants



AI is having real-world impact

- Public imagination
- Health



The Boston Globe, 2018

AI is having real-world impact

- Public imagination
- Health
- Science
 - Chemistry and biology

Science & technology | The 2024 Nobel prizes

AI wins big at the Nobels

Awards went to the discoverers of micro-RNA, pioneers of artificial-intelligence models and those using them for protein-structure prediction

The Economist, 2024

AI is having real-world impact

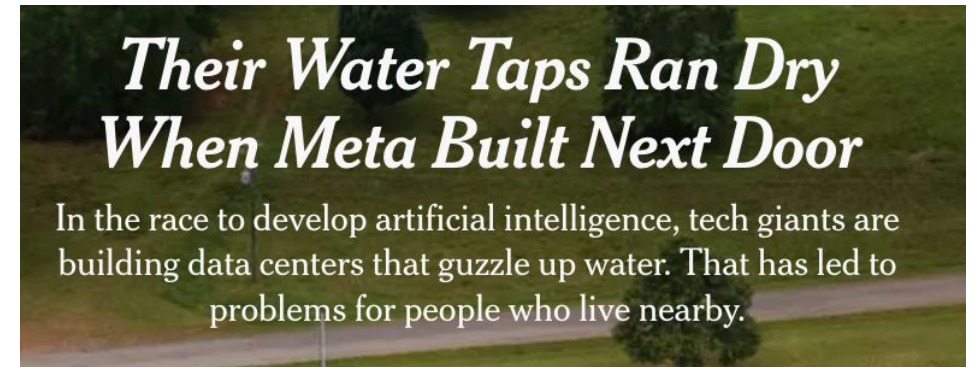
- Public imagination
- Health
- Science
 - Chemistry and biology
 - Physics



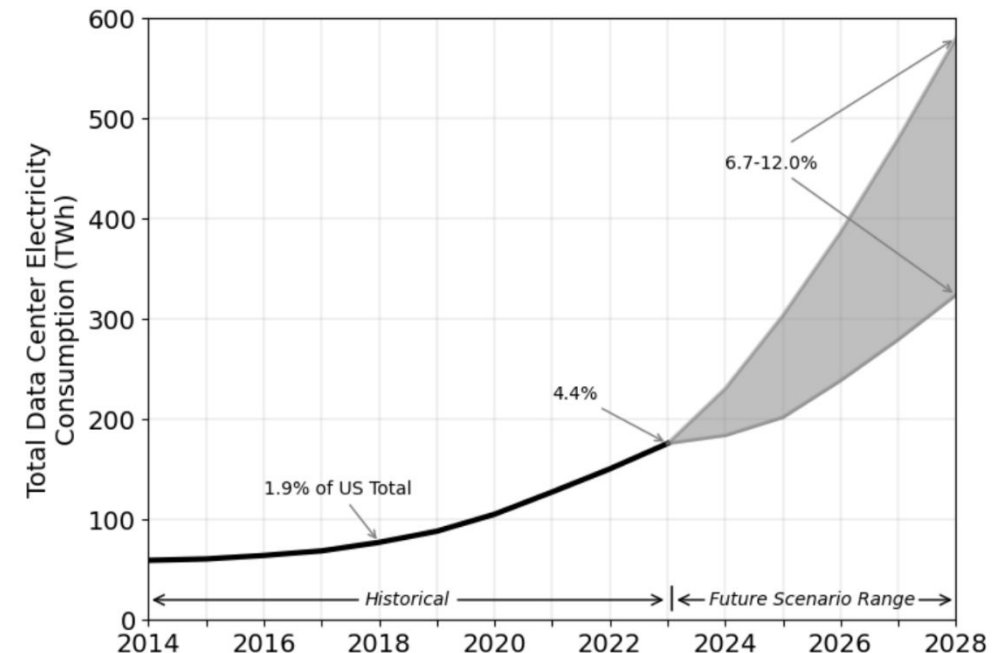
Wired, 2022

AI is having real-world impact

- Public imagination
- Health
- Science
 - Chemistry and biology
 - Physics
 - Resource use



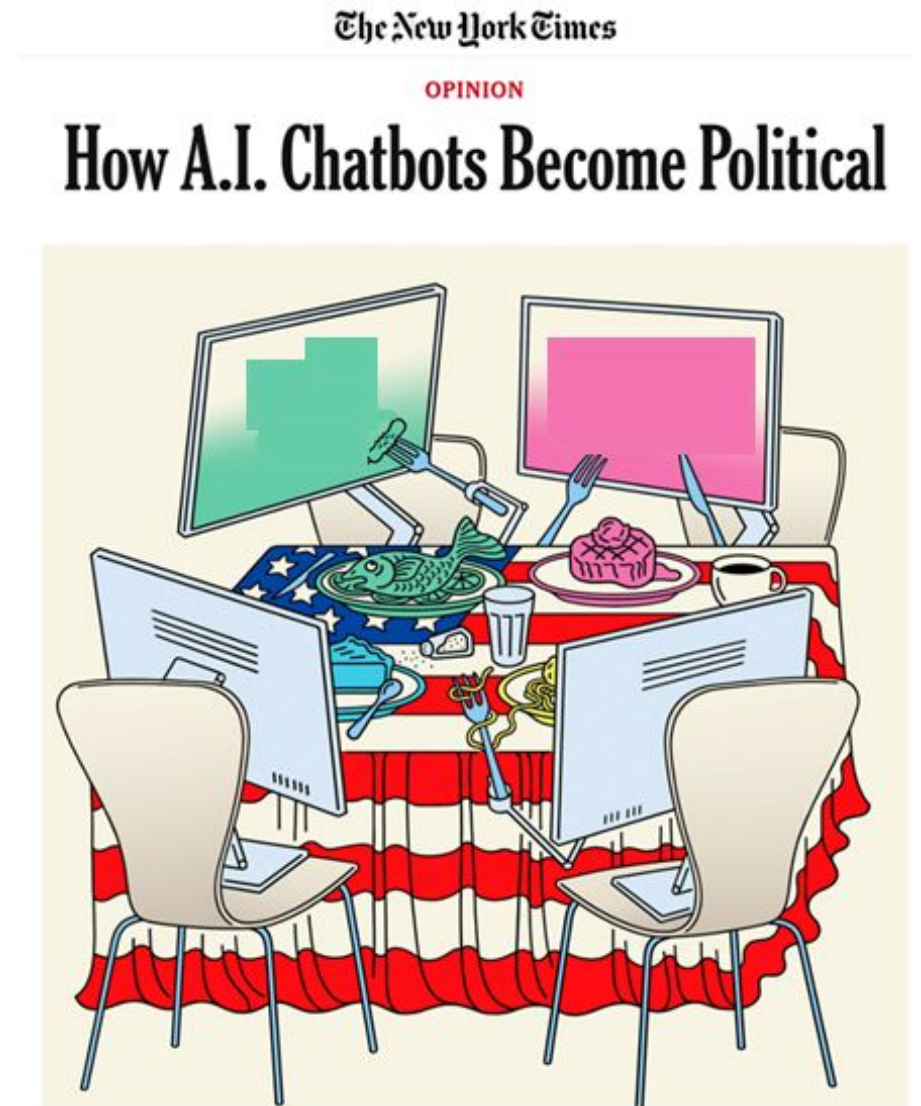
The New York Times, 2025



Lawrence Berkeley National Lab, 2024

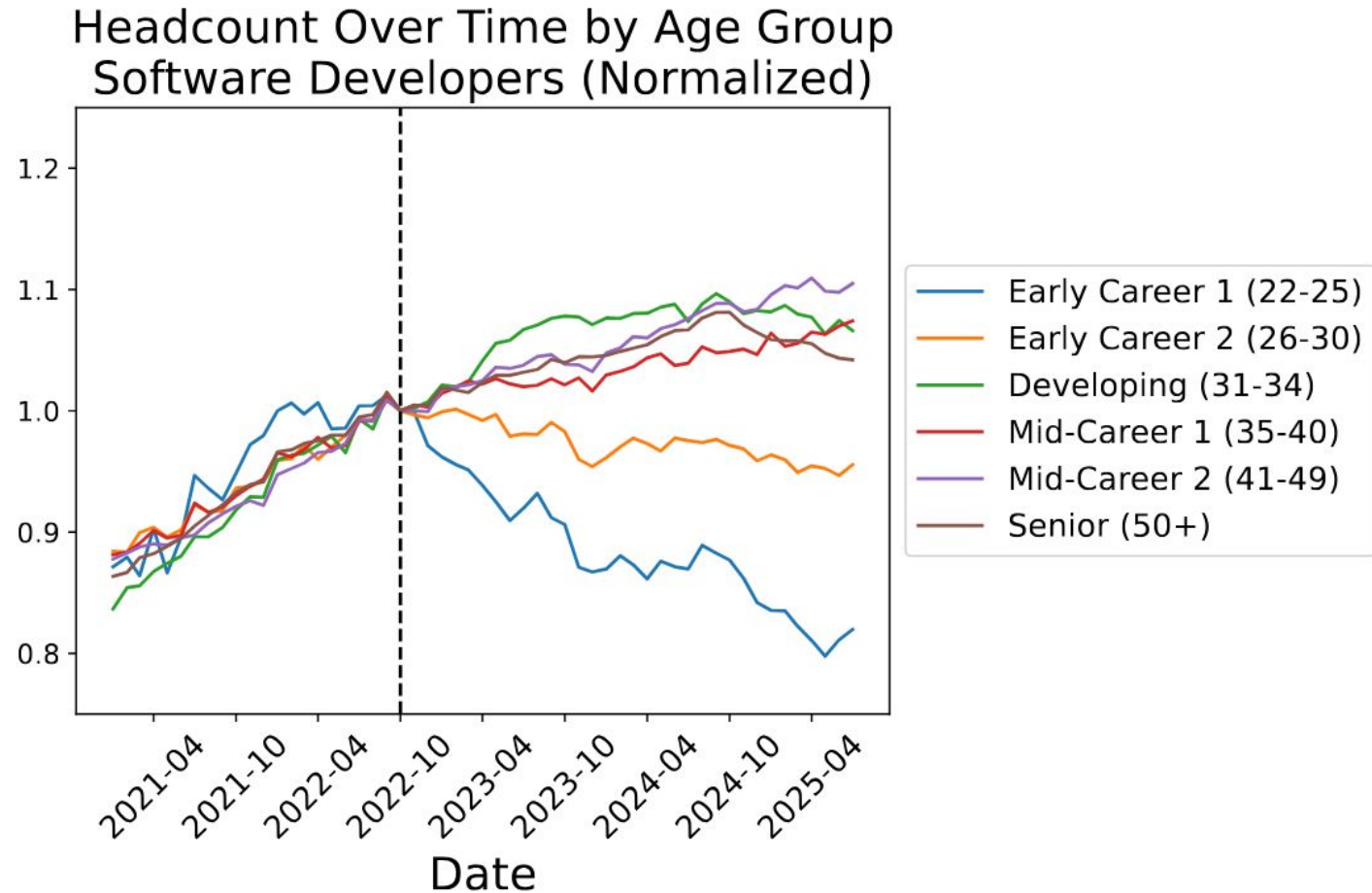
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AI is having real-world impact

- Public imagination
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AI is having real-world impact

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- Labor
- Education

BREAKING

ChatGPT In Schools: Here's Where It's Banned—And How It Could Potentially Help Students

Arianna Johnson Forbes Staff

I cover the latest trends in science, tech and healthcare.

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Jan 18, 2023, 02:31pm EST

Forbes, 2023

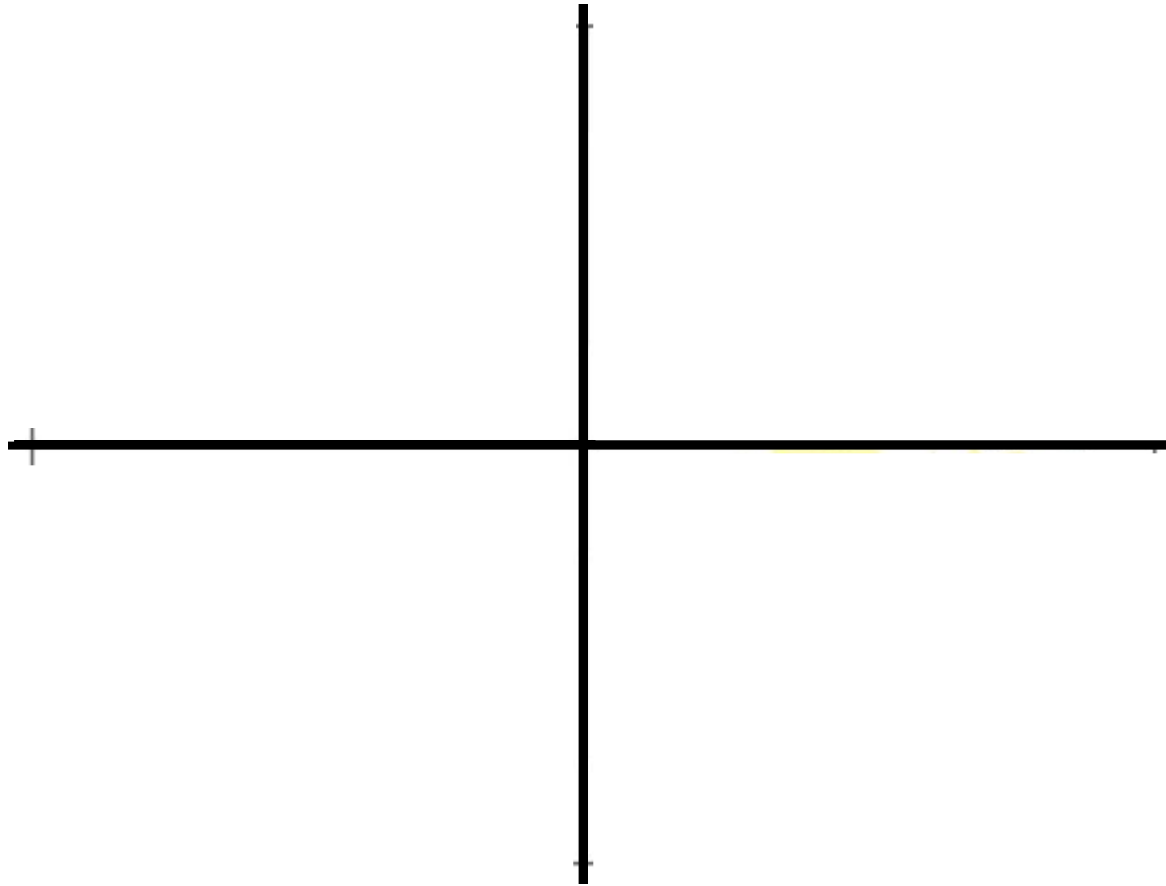
AI is having real-world impact

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- Education

Ok, but what actually is AI???

1.1 What is AI?

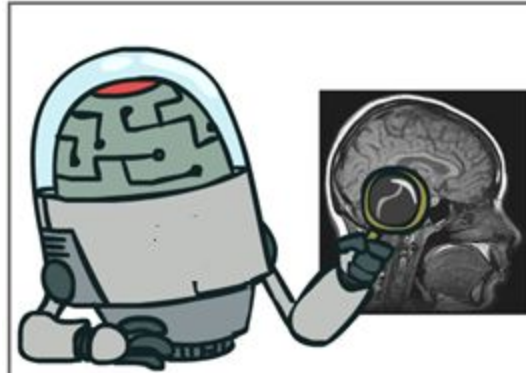
Should we make machines that...



1.1 What is AI?

Should we make machines that...

Think like people?



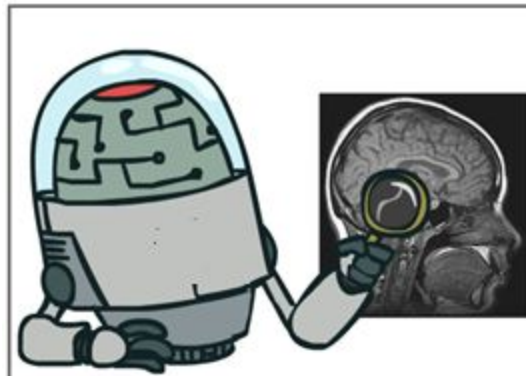
1.1.2: Thinking humanly

- We can learn about human thought in three ways:
 - **introspection**—trying to catch our own thoughts as they go by;
 - **psychological experiments**—observing a person in action;
 - **brain imaging**—observing the brain in action.
- Once we have a sufficiently precise theory of the mind, it becomes possible to express the theory as a computer program.
- The interdisciplinary field of **cognitive science** brings together computer models from AI and experimental Cognitive science techniques from psychology to construct precise and testable theories of the human mind.
- Real cognitive science, however, is necessarily based on experimental investigation of actual humans or animals.

1.1 What is AI?

Should we make machines that...

Think like people?



Act like people?



1.1.1: Acting humanly

- The Turing test approach

- natural language processing to communicate successfully in a human language;
- knowledge representation to store what it knows or hears;
- automated reasoning to answer questions and to draw new conclusions;
- machine learning to adapt to new circumstances and to detect and extrapolate patterns.

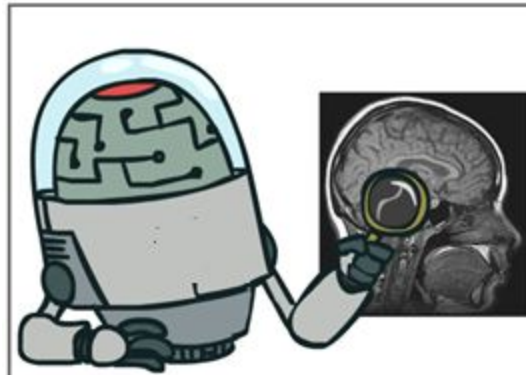
- Total Turing test

- computer vision and speech recognition to perceive the world;
 - robotics to manipulate objects and move about.
-
- However, it is more important to study the underlying principles of intelligence
 - “**artificial flight**” succeeded we stopped imitating birds and started using wind tunnels and learning about aerodynamics.
 - **Aeronautical engineering** do not define the goal “machines that fly so exactly like pigeons that they can fool even other pigeons.”

1.1 What is AI?

Should we make machines that...

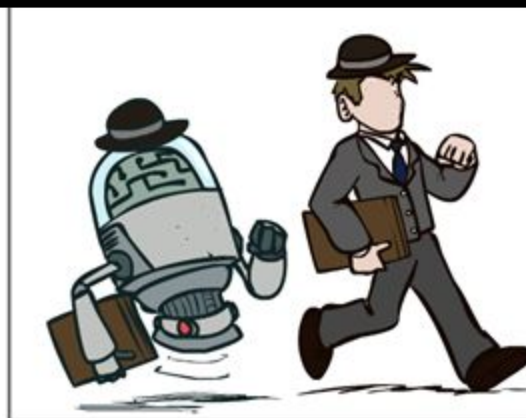
Think like people?



Think rationally?



Act like people?



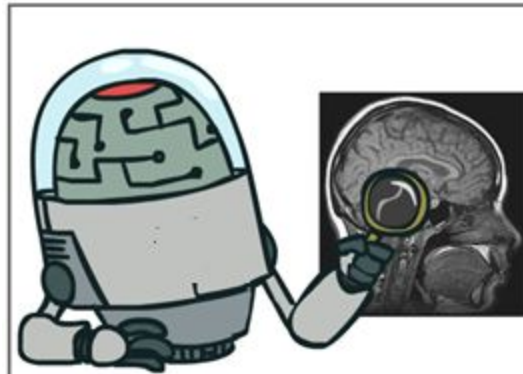
1.1.3: Thinking rationally

- The system design that think "correctly and logically"
 - Based on logic and mathematics
 - Focuses on correctness of reasoning
 - Uses rules like "if-then" logic
- Limitation
 - Real-world problems often involve "uncertainty and incomplete information", which pure logic cannot handle well.
 - We simply don't know the rules of, say, politics or warfare in the same way that we know the rules of chess or arithmetic.
- The theory of probability fills this gap, allowing rigorous reasoning with uncertain information.
- To generate intelligent behavior, theory of rational action is needed.

1.1 What is AI?

Should we make machines that...

Think like people?



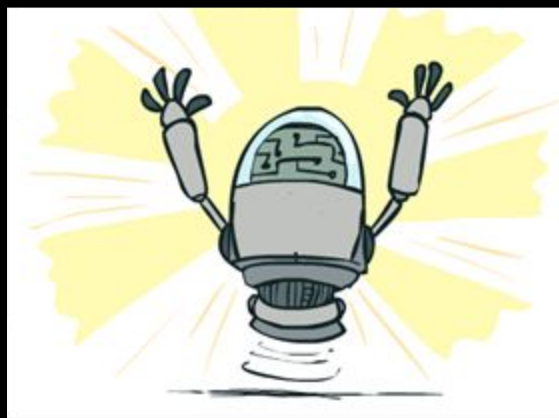
Think rationally?



Act like people?



Act rationally?

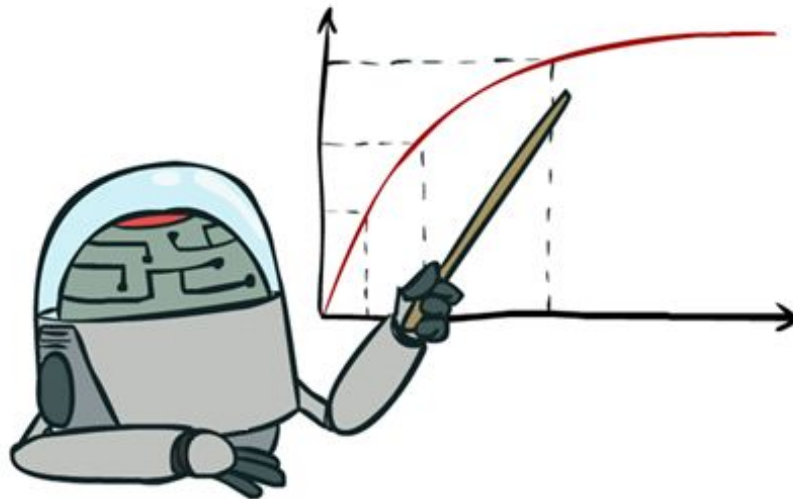


1.1.4: Acting rationally: The rational agent approach

- An AI system (called an agent) acts to achieve its goals in the best possible way
 - given its perceptions and
 - knowledge of the environment.
- Rational action does not mean perfect action, but optimal action under given constraints.
 - Focuses on outcomes rather than human similarity
 - Uses probability, decision theory, and learning
 - Works well in real-world, uncertain environments
- Example: A self-driving car choosing the safest and fastest route.
- Advantage: More practical and flexible than other approaches.

1.1.4: Rational Decisions

- We'll use the term **rational** in a very specific, technical way:
 - Rational: *maximally achieving pre-defined goals*
 - Goals are expressed in terms of the **utility** of outcomes
 - World is uncertain, so we'll use **expected** utility
 - Being rational means acting to **maximize your expected utility**



1.1.5: Beneficial machines

- Returning to the apparently **unproblematic** example of chess, consider what happens if the machine is intelligent enough to reason and **act beyond the confines of the chessboard**.
- In that case, it might attempt to increase its chances of winning by such ruses as:
 - hypnotizing or
 - blackmailing its opponent or
 - bribing the audience to make rustling noises during its opponent's thinking time.
 - place the board so the sun is in your opponent's eyes.
- These behaviors are not “unintelligent” or “insane”; they are a logical consequence of defining winning as the sole objective for the machine.

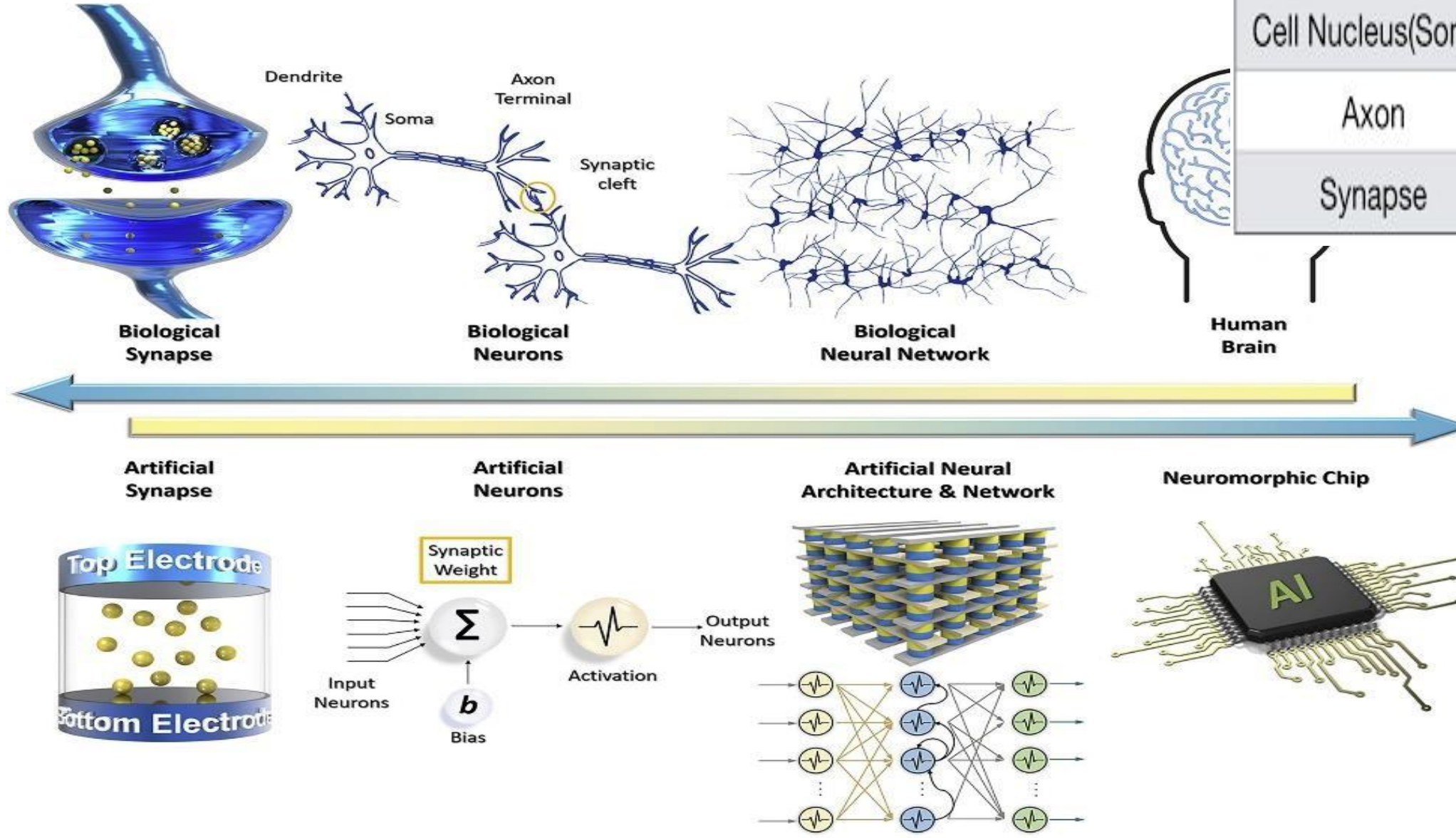
1.1.5: Beneficial machines

- It is impossible to anticipate all the ways in which a machine pursuing a fixed objective might misbehave.
- We don't want machines that are intelligent in the sense of pursuing **'their'** objectives; we want them to pursue **'our'** objectives.
 - A self-driving car, driving along any road incurs a risk of injury due to other errant drivers, equipment failure, and so on; thus, a strict goal of safety requires staying in the garage.
- When a machine knows that it doesn't know the complete objective, it has
 - to act cautiously,
 - to ask permission,
 - to learn more about our preferences through observation, and
 - to defer to human control.
- Ultimately, we want agents that are provably beneficial to humans

1.2: Foundations of Artificial Intelligence

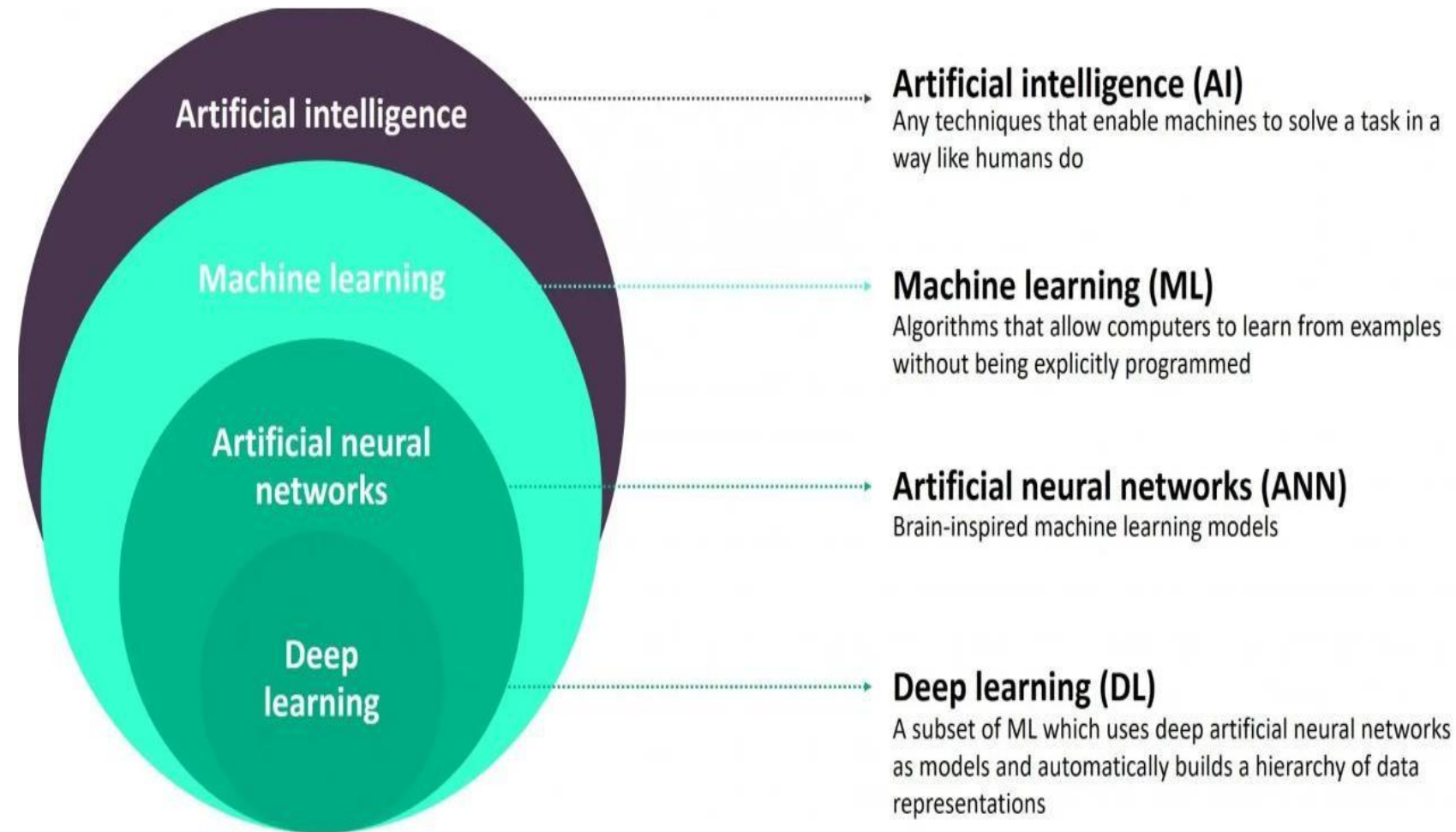
- **Philosophy** Gave AI the idea of logical reasoning and thinking rationally.
- **Mathematics** Provides logic, probability, and formulas used for reasoning and learning.
- **Economics** Helps AI choose the best action using limited resources wisely.
- **Neuroscience** Inspired brain-based models such as neural networks.
- **Psychology** Explains how humans think, learn, and solve problems.
- **Computer Engineering** Supplies hardware and computing power to run AI systems.
- **Control Theory & Cybernetics**
 - Helps machines in automation, stability, and control actions using feedback mechanisms.
- **Linguistics**
 - Enables AI to understand and generate human language.

Biological Neurons



Biological Neuron	Artificial Neuron
Dendrites	Input
Cell Nucleus(Soma)	Node
Axon	Output
Synapse	Interconnections

Biological Neurons



Biological Neuron	Artificial Neuron
Dendrites	Input
Cell Nucleus(Soma)	Node
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Synapse	Interconnections

What About the Brain?

- Brains (human minds) are very good at making rational decisions, but not perfect
- Brains aren't as modular as software, so hard to reverse engineer!
- AI may be better than brains at some tasks
- *“Brains are to intelligence as wings are to flight”*
- We can't yet build AI on the scale of the brain
 - ~100T synapses in the human brain vs ~1.8T weights in GPT4
- Still, the brain can be a great inspiration for AI! (Subhan Allah)

~1.8T weights:

Space: a large office floor

if 200KWh for a month=1home,

Power: 60GWh for an hour,

so 3 lack homes for a month.

720*3lack = 21 crore, 6 lack homes/ hour

Watch the videos please

- Google data center (11 years back)
- <https://www.youtube.com/watch?v=XZmGGAbHqa0&t=2s>
- How mind control Bionic arm (2 years back)
- https://www.youtube.com/watch?v=lpw_2A2T_wg
- How such complete our arms are. Subhan Allah
- Watch some recent videos on AI data center, to understand the differences with traditional data center.

1.3: A (Short) History of AI

- 1940-1950: Early days: neural and computer science meet
 - 1943: McCulloch & Pitts: Perceptron–boolean circuit model of brain
 - 1950: Turing's “Computing Machinery and Intelligence”
- 1950—70: Excitement! Logic-driven
 - 1950s: Early AI programs, including Samuel's checkers program, Newell & Simon's Logic Theorist, Gelernter's Geometry Engine
 - 1956: Dartmouth meeting: “Artificial Intelligence” adopted

*“We propose that a **2-month, 10-man study of artificial intelligence** be carried out **during the summer of 1956** at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that **every aspect of learning** or **any other feature of intelligence** can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. **We think that a significant advance can be made** in one or more of these problems if a carefully selected group of scientists work on it together for a summer.”*

1.3: A (Short) History of AI

- **1970—90: Knowledge-based approaches**
 - 1969—79: Early development of knowledge-based systems
 - 1980—88: Expert systems industry booms; backpropagation makes it feasible to train multi-layer neural networks
 - 1988—93: Expert systems industry busts: “AI Winter”
- **1990—2010: Statistical approaches, agents**
 - Resurgence of probability, focus on uncertainty
 - Agents and learning systems... “AI Spring”?
 - 1992: TD-Gammon achieves human-level play at backgammon
 - 1997: Deep Blue defeats Gary Kasparov at chess
 - 2002: Embodied AI; Roomba vacuum invented
- **2010—2017: Big Data, GPUs, Deep Learning**
 - 2011: Apple releases Siri
 - 2012: AlexNet wins ImageNet competition
 - 2015: DeepMind achieves “human-level” control in Atari games
 - 2016: DeepMind’s AlphaGo defeats Lee Sedol at Go
 - 2016: Google Translate migrates to neural networks

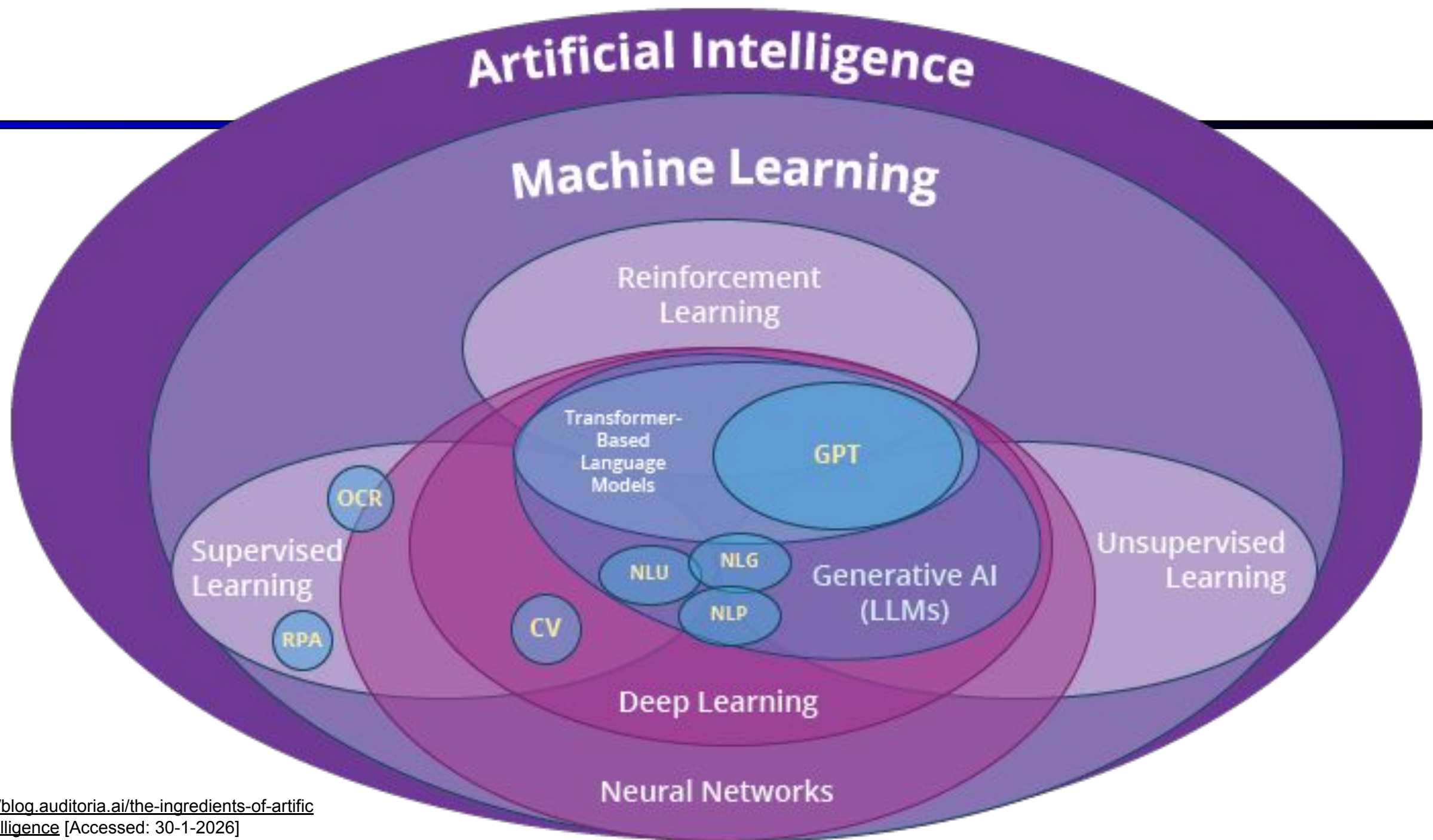
1.3: A (Short) History of AI

- 2017—: Scaling Up, Large Language Models
 - 2017: Google invents Transformer architecture
 - 2017: DeepStack/Libratus defeat humans at poker
 - 2018-2020: AlphaFold predicts protein structure from amino acids
 - 2021-2022: Modern text-to-image generation
 - 2022: OpenAI releases ChatGPT (Generative Pre-trained Transformer)
 - 2023: Every other company also releases a chatbot
 - 2024: Nobel prizes in physics and chemistry go to AI advances
 - 2025:Agentic AI

For example, an AI agent in medical imaging might be a CNN that classifies tumours in scans—nothing more.

In contrast, an agentic AI system would integrate multiple such agents (e.g., one fetching patient history, another cross-validating MRI and X-ray results, and a third alerting doctors) into a cohesive, goal-driven pipeline.

Criteria	Traditional AI	Agentic AI
Task assignment	solves task assigned by human	solves a task instead of human
Autonomy	follows predefined tasks	autonomous and able to make decisions
Learning capability	limited learning by training data	continuous and evolving
Behavior	reactive	proactive
Input dependency	manual input required	self initiating
Scope	single task focused	orchestrated workflows



This Course: Designing Rational Agents

- An **agent** is an entity that perceives and acts.
- A **rational agent** selects actions that maximize its (expected) **utility**.
- Characteristics of the **percepts**, **environment**, and **action space** dictate techniques for selecting rational actions
- This course is about:
 - General AI techniques for a variety of problem types
 - Learning to recognize when and how a new problem can be solved with an existing technique

